The Bilal Morning

Monday, 12 October 2026 BACKFILLED EDITION 12 stories · 15 sources

THREE THINGS TO KNOW

1. Space agencies and private aerospace companies increasingly rely on 3D printing to manufacture up to 85 percent of modern rocket engine components.
2. Air Force Secretary Troy Meink announced that the U.S. has deployed operational weapons to space, signaling an escalation in orbital militarization.
3. Researchers in China developed an innovative eggshell-inspired shield that significantly slows high-speed space debris during simulations.

Rocket Engines Are Now Mostly Built With Advanced 3D Printing

Space agencies and private aerospace companies increasingly rely on 3D printing to manufacture up to 85 percent of modern rocket engine components.

- Around 80 percent of parts in contemporary rocket engines are produced using 3D printing. [bgr.com]
  > "modern rocket engines include around 80% 3D-printed parts."
- NASA and Elementum 3D developed a specialized aluminum alloy that provides strength and heat resistance while cutting weight. [bgr.com]
  > "The result was a new aluminum alloy that is strong and heat-resistant without weighing down rockets."
- SpaceX utilized custom alloy printing to minimize welds and accelerate engine fabrication. [bgr.com]
  > "SpaceX, for example, created a nickel-chromium alloy called Inconel for its early SuperDraco engines."
- Up to 20 percent of engine parts still require conventional fabrication to handle conditions beyond the limits of additive manufacturing. [bgr.com]
  > "However, that still leaves 15-20% of parts at minimum for traditional manufacturing."

In the image: A rocket lifts off from a launch pad framed by tall lightning towers and ground radar domes.

✓ 4 of 4 claims verified against sources · 1 source

bgr.com

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SPACE EXPLORATION

U.S. Confirms First-Time Deployment of Operational Weapons in Space

Air Force Secretary Troy Meink announced that the U.S. has deployed operational weapons to space, signaling an escalation in orbital militarization.

- Air Force Secretary Troy Meink publicly confirmed that the military has deployed operational weapons in Earth's orbit.

  [opednews.com]

  > "Air Force Secretary Troy Meink became the first U.S. official to publicly announce that the military has deployed weapons to space"

- The deployment aligns with President Trump's focus on space, including the proposed 'Golden Dome' missile defense system.

  [opednews.com]

  > "In his second term, he has announced the 'Golden Dome' missile defense system, which includes plans to put U.S. weapons in space."

- Defense officials emphasized the necessity of achieving complete dominance and avoiding orbital parity with adversaries. [opednews.com]

  > "We cannot afford to have a fair fight in space-- I'm not interested in orbital parity."

- Top military adviser Gen. Dan Caine urged troops to prepare for combat extending from the seabed to cislunar space.

  [opednews.com]

  > "is prepared to fight, endure and win ... from the seabed to cislunar space"

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opednews.com

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Egg-Inspired Spacecraft Shield Outperforms Traditional Aluminum Protection

Researchers in China developed an innovative eggshell-inspired shield that significantly slows high-speed space debris during simulations.

- Scientists in China created a new spacecraft shield inspired by the natural geometry of eggshells.

  [dailygalaxy.com]

  > "Scientists in China have developed an egg-inspired spacecraft shield that reduced a simulated debris projectile's speed by nearly 65%"

- The design uses hollow aluminum shells filled with water arranged between metal plates. [dailygalaxy.com]

  > "The shells were filled with water and arranged between protective metal plates."

- Simulations showed the egg-inspired shield reduced projectile speed by nearly 65 percent, outperforming a standard aluminum plate.

  [dailygalaxy.com]

  > "the projectile lost 64.9% of its initial speed, compared with a 51% reduction when passing through a conventional aluminum plate."

In the image: The image illustrates Earth surrounded by an orbit filled with various pieces of space debris and fragmented satellites, a visual context not described in detail by the text.

✓ 3 of 3 claims verified against sources · 1 source

dailygalaxy.com

You engaged with 2 of 2 Space exploration stories you saw

NASA Astronaut Photographs Moon-Like Rock Formation in Madagascar

An International Space Station astronaut captured an ancient anorthosite formation in Madagascar that mirrors the Moon's bright highlands.

- An astronaut aboard the International Space Station photographed a lunar-like rock formation in Madagascar made of anorthosite. [sciencedaily.com]

  > "An astronaut aboard the International Space Station photographed a remarkable formation in Madagascar that looks surprisingly similar to the Moon's bright highlands."

- The Saririaky anorthosite massif covers approximately 100 square kilometers and dates back at least 600 million years. [sciencedaily.com]

  > "Geological research suggests that the Saririaky formation originated during the late Precambrian, at least 600 million years ago."

- Anorthosite is an intrusive igneous rock that makes up much of the Moon's reflective highlands. [sciencedaily.com]

  > "The rock, called anorthosite, is an igneous material that makes up much of the Moon's bright, reflective highlands."

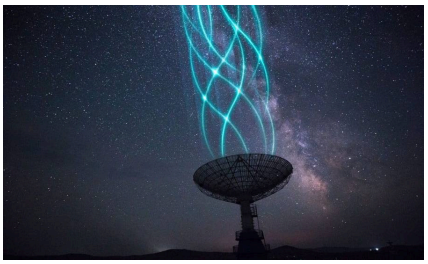
- The surrounding region experienced intense deformation and ancient continental collisions during the formation of Gondwana. [sciencedaily.com]

  > "Scientists believe much of this transformation occurred as ancient landmasses collided during the formation of Gondwana"

✓ 4 of 4 claims verified against sources · 1 source

sciencedaily.com

You engaged with 2 of 2 Space exploration stories you saw



Astronomers Discover Extremely Powerful Radio Galaxy From the Early Universe

Researchers have identified TXS 2354+015, a massive and exceptionally powerful radio galaxy that existed nearly 12.5 billion years ago.

- TXS 2354+015 existed when the universe was under 1.2 billion years old and has been identified as the second-most-distant radio galaxy ever found.

  [dailygalaxy.com]

  > “place TXS 2354+015 at a redshift of 4.946, making it the second-most-distant radio galaxy ever found”

- The discovery was led by Barbara Balmaverde's team by combining optical and radio observations to find distant objects. [dailygalaxy.com]

  > “The galaxy was identified by a team led by Barbara Balmaverde from the INAF Astrophysical Observatory of Turin”

- Calculations of the galaxy's radio energy suggest it has a higher estimated intrinsic radio luminosity than any previously known high-redshift radio galaxy.

  [dailygalaxy.com]

  > “TXS 2354+015 has a higher estimated intrinsic radio luminosity at 500 MHz than any previously known high-redshift radio galaxy”

- Researchers estimate the host galaxy could contain around 2 trillion solar masses in stars, although this figure carries major uncertainties. [dailygalaxy.com]

  > “its host could contain around 2 trillion solar masses in stars, although they stressed that this figure comes with major uncertainties”

In the image: The image conceptually visualizes a radio telescope emitting signal waves toward deep space, which is not described in the text.

✓ 4 of 4 claims verified against sources · 1 source

dailygalaxy.com

You engaged with 2 of 3 Astronomy and astrophysics stories you saw

Inverclyde Teacher Finds Comfort in Astronomy and Fond Childhood Memories

After early retirement due to ill-health, Marion McNeil founded a local astronomy group that has since been nationally recognised.

- Marion McNeil was forced into early retirement due to a period of ill-health, which prompted her to rediscover her love of the stars.

  [greenocktelegraph.co.uk]

  > “WHEN a period of ill-health forced Inverclyde teacher Marion McNeil into early retirement she rediscovered her love of the stars”

- She founded the Inverclyde Sky Watchers in 2015 after attending a beginner's astronomy course during her rehabilitation.

  [greenocktelegraph.co.uk]

  > “The 74-year-old founded Inverclyde Sky Watchers in 2015 and has even been nationally recognised for her work with the group.”

- Her passion for the night sky stems from her childhood on Islay, where her German mother shared stories with her. [greenocktelegraph.co.uk]

  > “She has loved looking at the stars ever since she was a child growing up on Islay when her German mother shared stories”

- Marion McNeil was honored with the British Astronomical Association's Sir Patrick Moore Prize for her contributions.

  [greenocktelegraph.co.uk]

> “She then won the British Association’s Sir Patrick Moore Prize, without even knowing she had been nominated.”

In the image: The image provides a portrait of Marion McNeil, showing her smiling while wearing a dark denim-style shirt and pearl earrings, which is not captured in the text.

✓ 4 of 4 claims verified against sources · 1 source

greenocktelegraph.co.uk

You engaged with 2 of 3 Astronomy and astrophysics stories you saw

Physics constants may be finely tuned to make life possible

Scientists discover that tiny changes to fundamental physical constants could alter liquid viscosity and prevent essential biological processes from occurring.

- Physicist Kostya Trachenko discovered that physical constants must allow liquids to support life. [sciencedaily.com]

  > “Their findings suggest that even relatively small changes to some of the universe's most basic physical constants could dramatically alter how liquids flow”

- Fundamental constants determine the lower limit of liquid viscosity.

  [sciencedaily.com]

  > “fundamental physical constants place a lower limit on the viscosity of liquids”

- Cellular life depends heavily on fluid movement and viscosity. [sciencedaily.com]

  > “Life processes in and between living cells require motion and it is viscosity that sets the properties of this motion”

- Even a minor percentage change in fundamental constants would disrupt biological functions.

  [sciencedaily.com]

  > “viscosity of our blood would become too thick or too thin for body functioning with only a few per cent change”

✓ 4 of 4 claims verified against sources · 1 source

sciencedaily.com

You engaged with 2 of 3 Astronomy and astrophysics stories you saw

A Guide to Observing and Imaging the Ancient Hercules Cluster

Discover the history, age, and unique characteristics of the Hercules Cluster, an ancient celestial target ideal for quick astronomical viewing.

- Globular clusters like the Hercules cluster require only about ten minutes of imaging time to capture a good picture.

  [greenocktelegraph.co.uk]

  > “For globular clusters, I find that ten minutes is usually more than plenty to get you a pretty picture.”

- The Hercules cluster consists of around 300,000 stars packed into a diameter of about 145 light years.

  [greenocktelegraph.co.uk]

  > “most of the 300,000 stars within the Hercules cluster are contained within its diameter of about 145 light years.”

- Edmond Halley discovered the Hercules cluster in 1714 and described it as a small patch visible to the naked eye under good conditions.

  [greenocktelegraph.co.uk]

  > “The Hercules cluster was discovered in 1714 by the renowned astronomer Edmond Halley.”

- At 12 to 14 billion years old, the Hercules cluster contains some of the oldest surviving stars in the universe. [greenocktelegraph.co.uk]

  > “At between 12 and 14 billion years old, it is almost as old as the Universe itself.”

- In 1974, astronomers broadcast a binary coded radio signal from the Arecibo telescope toward the Hercules cluster.

[greenocktelegraph.co.uk]

> "In 1974 when they decided to hail other potential extra-terrestrial intelligences out there with a radio signal."

✓ 5 of 5 claims verified against sources · 1 source

greenocktelegraph.co.uk

You engaged with 2 of 3 Astronomy | *and astrophysics stories you saw*

CRICKET



In the image: The image shows Georgia Voll celebrating her century with her bat and helmet raised on the field.

✓ 3 of 3 claims verified against sources · 1 source

theguardian.com

You engaged with 0 of 1 Cricket stories you saw

Georgia Voll's Century Secures Australia ODI Series Win

Georgia Voll smashed an unbeaten century to lead Australia to a comfortable nine-wicket victory over Bangladesh and a 2-0 series lead.

- Georgia Voll scored an unbeaten century in only her 15th match to help Australia secure a comfortable win against Bangladesh. [theguardian.com]

  > "Opener Georgia Voll smashed her third ODI century in just her 15th match to secure Australia a comfortable nine-wicket win over Bangladesh"

- Sobhana Mostary and Nigar Sultana formed a record-breaking partnership for Bangladesh. [theguardian.com]

  > "Mostary, 24, made a magnificent 118 off 134 deliveries and with her captain Nigar Sultana (74no off 114) added 195 for the fourth wicket"

- Voll set a new venue record for the highest individual women's ODI total at Allan Border Field. [theguardian.com]

  > "Voll's unbeaten 131 off 97 deliveries became the highest individual women's ODI total at the venue, breaking a record Mostary had equalled only hours earlier."



Pakistan Imposes Emergency Rule in Khyber Pakhtunkhwa Province

Pakistan's president has suspended the provincial government of Khyber Pakhtunkhwa, led by Imran Khan's party, amid escalating political confrontation and planned protest marches.

- Pakistan's president declared a state of emergency and suspended the government of Khyber Pakhtunkhwa province, which is ruled by Imran Khan's party. [straitstimes.com]

  > “Pakistan's president suspended the government in a province ruled by the party of jailed former Prime Minister Imran Khan after declaring a state of emergency on Sunday”

- The federal government justified the takeover by claiming the provincial administration failed to address terrorism and militant threats. [straitstimes.com][aninews.in]

  > “accused the PTI-led leadership of deliberate indifference in the war against terrorism”

- PTI officials condemned the decision, warning that it would lead to heightened instability and further bloodshed. [straitstimes.com]

  > “This will only lead to further bloodshed, instability and collateral damage”

- The political crisis intensified as the PTI organized a march on Islamabad to demand the release of jailed former Prime Minister Imran Khan. [aninews.in][dailymirror.lk]

  > “The march, organised to demand the release of PTI founder Imran Khan, emerged as a major point of confrontation”

In the image: A heavily armed security officer stands guard behind police tape and container barricades set up to control access during the unrest.

✓ 4 of 4 claims verified against sources · 4 sources

straitstimes.com · moneycontrol.com · aninews.in · dailymirror.lk

You engaged with 0 of 2 Pakistan cricket team stories you saw



Verstappen on Pole for Singapore Grand Prix as Ferrari Pursues

Max Verstappen starts on pole in Singapore, but pressure from Ferrari, wet weather threats, and strategic challenges loom large at the Marina Bay Circuit.

- Max Verstappen has secured pole position for the Singapore Grand Prix on the Marina Bay Circuit. [grandprix247.com]

  > “Max Verstappen is on pole for the Singapore Grand Prix, exactly where he needs to be on the narrow, twisting Marina Bay Circuit.”

- Isack Hadjar experienced a mechanical issue during qualifying that left him out of contention. [grandprix247.com]

  > “Isack Hadjar suffered a mechanical failure, putting him out of contention in qualifying.”

- George Russell incurred a 40-place penalty for engine changes following a difficult sprint race. [grandprix247.com]

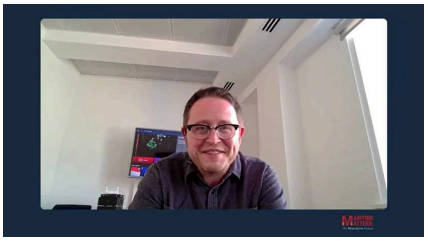
  > “After bottling the Sprint by spinning from the lead, George Russell has now taken a 40-place penalty for engine changes.”

In the image: The image shows the complete starting grid lineup and driver numbers for the Singapore Grand Prix against a nighttime city backdrop.

✓ 3 of 3 claims verified against sources · 1 source

grandprix247.com

Exploring: a Motorsport story outside your usual topics, to learn whether you want more



Jamie Airs Discusses Connectivity and Crew Welfare on Maritime Matters Podcast

KVH Industries' Global Content Sales Manager explains how digital solutions improve seafarer retention, entertainment, and cybersecurity.

- Crew welfare remains a primary focus for shipowners due to ongoing recruitment and retention hurdles. [marinelink.com]

  > “Crew welfare is a central theme to most if not all shipowner agendas, as recruiting and retaining seafarers remains historically challenging.”

- Modern crews expect digital connectivity that provides entertaining content and links them to shipmates. [marinelink.com]

  > “the question now turns from simple connectivity to quality content that now only help to keep seafarers entertained, but connected too to their shipmates.”

- Jamie Airs, Global Content Sales Manager at KVH Industries, addressed crew connectivity challenges in a recent podcast. [marinelink.com]

  > “Jamie Airs, Global Content Sales Manager, KVH Industries, talks through the challenges as well as the multitude of solutions from KVH”

In the image: The image shows Jamie Airs participating in a video call in an office setting, which is not mentioned in the text.

✓ 3 of 3 claims verified against sources · 1 source

marinelink.com

Exploring: a Society story outside your usual topics, to learn whether you want more

You're caught up.

That's today's paper. Your next edition arrives tomorrow morning.